

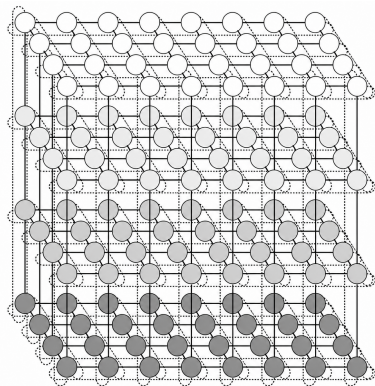
Beyond Half Twists: Joint Topology and Adaptive Routing Design for Asymmetric 3D Torus

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Motivation: asymmetric 3D torus

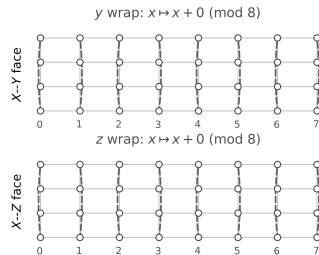
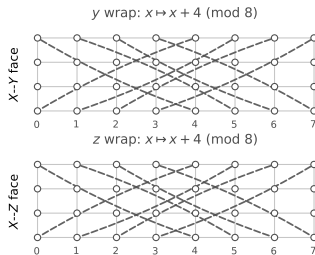
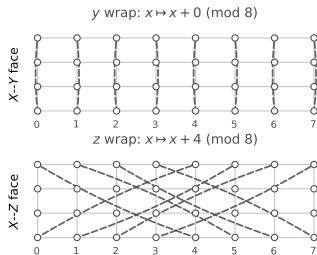
- ▶ A 3D torus offers a scalable degree-6 interconnect.
- ▶ Real deployments can be **asymmetric**: Google Cloud TPU v4 supports $8 \times 4 \times 4$ and $8 \times 8 \times 4$ torus slices.
- ▶ In $8 \times 4 \times 4$, packets travel farther along X : X links can bottleneck while Y/Z links are underused.



Unfolded $8 \times 4 \times 4$ torus

Prior idea: Twist 3D Torus

Twist a short-dimension wraparound link so that crossing the boundary also moves the packet in X . Consider a $k \times k \times 2k$ torus.



Definition 1. A Prismatic Twisted Torus (PTT) uses one half shift: $(p, q) = (0, k)$.

Definition 2. A Prismatic Doubly Twisted Torus (PDTT) uses two half shifts: $(p, q) = (k, k)$.

Definition 3. An ordinary 3D torus applies no boundary shift: $(p, q) = (0, 0)$.

Problem 1: Can we find a twist scheme that further reduces the distance?

Theorem 1

Among all dual twists (p, q) of a $2k \times k \times k$ torus,

$$\min_{p,q} D(T(2k, k, k; p, q)) = \left\lceil \frac{3k}{2} \right\rceil - 1.$$

The shift $(p, q) = (k - 1, k)$ attains this bound.

The derivation is non-trivial; interested readers may refer to the proof in our report.

Here, we only show how to calculate the average shortest distance for a given torus size (next page).

Problem 1: Can we find a twist scheme that further reduces the distance?

Twist	Diameter	Average shortest distance
(0, 0)	8	4.0000
(4, 0)	6	3.6250
(4, 4)	6	3.4375
(3, 4)	5	3.3750

How is average distance calculated?

For the ordinary torus, dimensions are independent, so $\bar{d} = 8/4 + 4/4 + 4/4 = 4$.

For a twisted torus, e.g., (3, 4):

$$(N_0, \dots, N_5) = (1, 6, 18, 38, 49, 16),$$
$$\bar{d} = \frac{1(6) + 2(18) + 3(38) + 4(49) + 5(16)}{128} = \frac{432}{128} = 3.3750.$$

Problem 2: does the shortest topology always win?

Let $m(s, t)$ be the number of distinct shortest paths from source s to destination t .

Twist	$\Pr[m(s, t) > 1]$	Mean $m(s, t)$	Maximum
(0, 0)	0.9213	81.1969	3360
(4, 0)	0.8898	27.9370	360
(4, 4)	0.8583	20.0315	360
(3, 4)	0.8583	9.7165	60

Key observation

Shorter paths, but fewer ways around congestion.

We will solve this problem through routing algorithm design.

Routing 1: AdaptiveRoute

1. Minimal candidates

$$M(v, t) = \{u \sim v \mid d(u, t) = d(v, t) - 1\}.$$

2. Adaptive selection

- ▶ Sum downstream credits over adaptive VCs.
- ▶ Choose the maximum; break ties randomly.

3. Escape fallback

- ▶ If no adaptive VC is available, recompute a spanning-tree output.
- ▶ Adaptive $\rightarrow$ Escape is allowed; Escape $\nrightarrow$ Adaptive.

Deadlock freedom

The dedicated escape VC follows only

$$\text{tree-up}^* \rightarrow \text{tree-down}^*.$$

Its channel-dependency graph is acyclic and remains reachable from the adaptive class.

Minimal + non-deterministic + deadlock-free

Routing 2: FlexRoute

Lateral candidate: an equal-distance neighbor

$$S(v, t) = \{u \sim v \mid d(u, t) = d(v, t)\}.$$

Each packet may use **at most one** lateral hop. It is considered only when minimal outputs have no free adaptive VC or are under credit pressure.

Two-hop bottleneck

$$B(u, t) = \min \left\{ C(v, u), \max_{w \in M(u, t)} C(u, w) \right\}.$$

Here, $C(a, b)$ is the total number of downstream credits on the adaptive VCs of the link from router a to its neighbor b .

Score

$$\text{score}(u) = B(u, t) + \alpha |M(u, t)|.$$

Why it is safe

One-hop slack prevents livelock.
Deadlock freedom still comes from the same dedicated, irreversible escape subnetwork.

If none qualifies, invoke
AdaptiveRoute unchanged.

Topology implementation

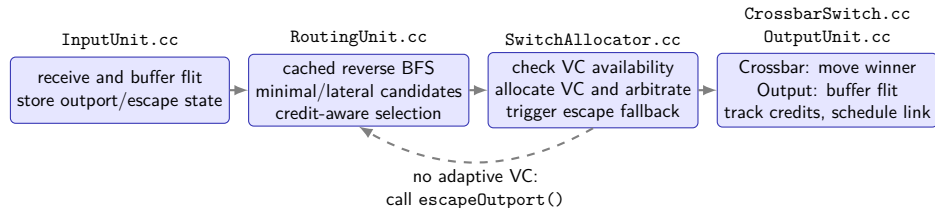
- ▶ TwistedTorusGraph.py
 - ▶ $ID \leftrightarrow (x, y, z)$
 - ▶ six neighbors and boundary shifts
- ▶ TwistedTorus.py
 - ▶ one router per coordinate
 - ▶ controller ExtLinks
 - ▶ six directed IntLinks/router
- ▶ Five modes: none, single, double, synth, custom.

Python topology construction and C++ routing use the same neighbor definition.

```
1 east = ((x + 1) % size_x, y, z)
2 west = ((x - 1) % size_x, y, z)
3
4 if y == size_y - 1 and twist_mode != "none":
5     north = ((x + y_shift) % size_x, 0, z)
6 else:
7     north = (x, (y + 1) % size_y, z)
8 if y == 0 and twist_mode != "none":
9     south = ((x - y_shift) % size_x, size_y - 1, z)
10 else:
11     south = (x, (y - 1) % size_y, z)
12
13 if z == size_z - 1 and twist_mode in ("double", "synth", "custom"):
14     up = ((x + z_shift) % size_x, y, 0)
15 else:
16     up = (x, y, (z + 1) % size_z)
17 if z == 0 and twist_mode in ("double", "synth", "custom"):
18     down = ((x - z_shift) % size_x, y, size_z - 1)
19 else:
20     down = (x, y, (z - 1) % size_z)
21
22 destinations = (east, west, north, south, up, down)
```

Ordinary and twisted wraparound links

Routing implementation



► Eligibility in flexOutput()

$$B_M = \max_{x \in M(v,t)} B(x,t),$$

$$Q_1 = (B_M < \tau C_{\max}),$$

$$Q_2(u) = (B(u,t) \geq B_M + H),$$

$$P(u) = \neg F_M \vee (Q_1 \wedge Q_2(u)).$$

A lateral candidate also needs unused slack, a free adaptive VC, and a true pressure test.

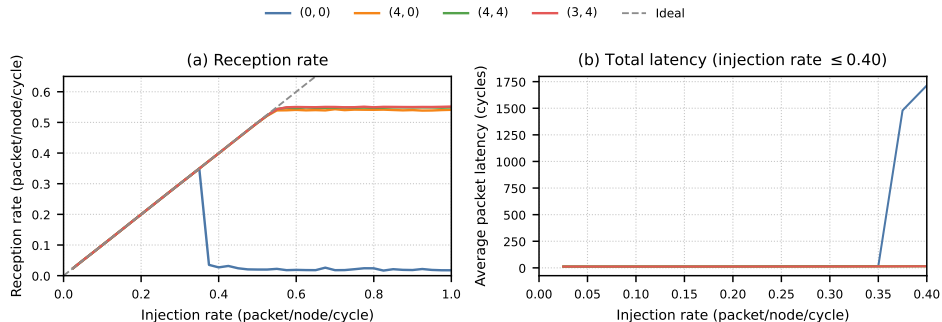
► Near-best randomized selection

$$E_\epsilon = \{u : \text{score}_{\max} - \text{score}(u) \leq \epsilon\},$$

$$W(u) = \max\{1, \text{score}(u) + 1\},$$

$$\Pr(u) = \frac{W(u)}{\sum_{x \in E} W(x)}, \quad E = E_\epsilon.$$

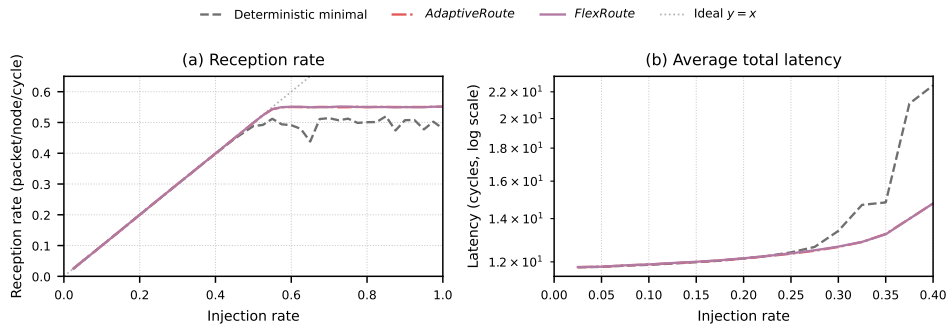
Topology



Twist	Peak recv. rate	Latency at 0.05
(0, 0)	0.349	13.0863
(4, 0)	0.543	12.3072
(4, 4)	0.549	11.9173
(3, 4)	0.552	11.7939

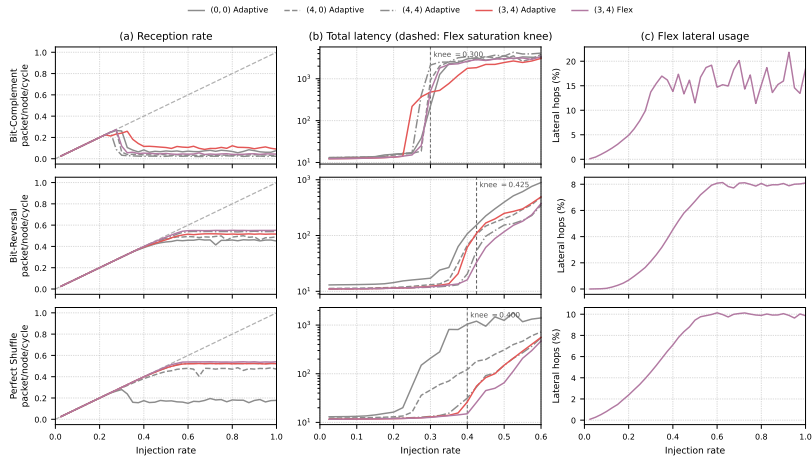
- Setup: $8 \times 4 \times 4$, VNet 0, one-flit packets, and 4 VCs/VNet (3 adaptive + 1 escape); FlexRoute uses $\tau = 100\%$, $H = 2$, and $\alpha = \epsilon = 1$.
- (3, 4) achieves the smallest low-load latency and the highest measured peak reception rate.

Routing



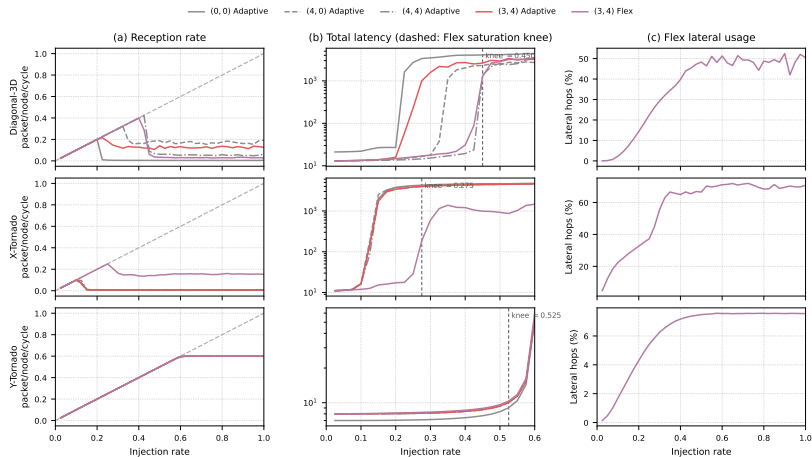
- ▶ AdaptiveRoute and FlexRoute reduce near-saturation latency relative to fixed-priority deterministic minimal routing.
- ▶ Under uniform random, FlexRoute stays close to AdaptiveRoute because lateral detours are rarely needed. Next, we showcase the power of FlexRoute.

FlexRoute unleashes the power of the (3,4) twist: permutation traffic



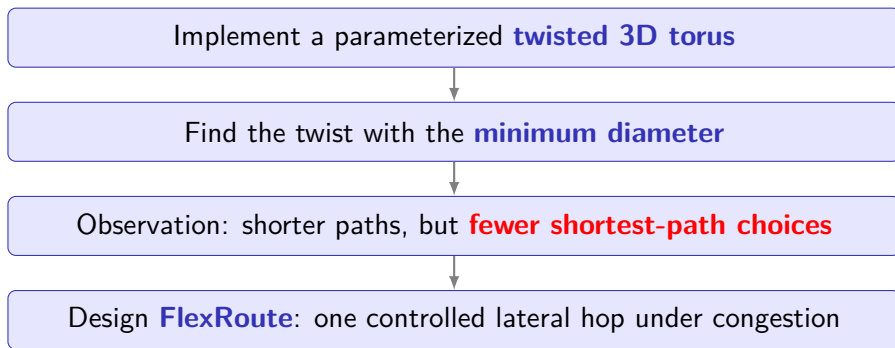
- ▶ Three permutation patterns evaluated in the original PTT/PDPT paper.
- ▶ For Bit-Reversal and Perfect Shuffle, (3,4) Flex achieves the best results.
- ▶ Under Bit-Complement, (3,4) Adaptive is not the best, but high lateral usage lets FlexRoute recover performance to the (4,4) level.

FlexRoute unleashes the power of the (3,4) twist: directional traffic



- X-Tornado:
FlexRoute shows its clearest throughput/latency gain and highest lateral usage.
- Diagonal-3D:
FlexRoute restores the (3,4) twist to a level comparable with (4,4).

From shorter topology to better routing



Topology and routing should be designed together.

Division of labor and takeaway

Fan Zhengyang

- ▶ topology design, theory, and implementation;
- ▶ most routing design and implementation;
- ▶ evaluation and figures;
- ▶ report and presentation.

Xu Yudong

- ▶ Part of routing implementation.
- ▶ Part of the report.

Takeaway

1. $(k - 1, k)$ minimizes diameter for $2k \times k \times k$ dual twists.
2. Shorter paths can reduce minimal-path diversity.
3. FlexRoute recovers useful path choices with one congestion-aware lateral hop and a deadlock-free escape VC.